

Project Report

Pharmaceutical Supply Chain Optimization

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Tools Used: Python (Pandas, Matplotlib, Seaborn) | MySQL | Power BI | Excel

1. Executive Summary

This project analyzes 100,000 pharmaceutical inventory records covering four drugs — Metformin, Lisinopril, Insulin, and Atorvastatin — to evaluate how well forecasted demand is being matched to planned stock levels, and whether the choice of restocking cadence (Weekly, Monthly, Quarterly) affects shortage risk. The analysis pipeline spans Python (cleaning and EDA), MySQL (business-question querying), and Power BI (interactive dashboarding). The headline finding is that the portfolio is overstocked in aggregate — stock exceeds demand by roughly 78 million units — yet 41.9% of individual Stock Keeping Unit records are still understocked, and this risk is essentially unchanged across all three restocking strategies. This points to a forecasting/allocation problem rather than a cadence problem.

2. Objective

To quantify understock and overstock exposure across the drug portfolio, test whether restocking strategy affects shortage risk, identify the highest-risk drugs and Stock Keeping Units, and package the findings into a reusable SQL + Python + Power BI analysis stakeholders can act on.

3. Dataset Overview

Attribute	Details
Source file	Pharmaceutical Supply Chain Optimization.xlsx
Records	100,000 rows (99,995 after cleaning)
Columns	Drug, Demand_Forecast, Optimal_Stock_Level, Restocking_Strategy
Drugs covered	Metformin, Lisinopril, Insulin, Atorvastatin
Restocking strategies	Weekly, Monthly, Quarterly
Demand range	1,000 – 9,999 units per record

4. Methodology

4.1 Data Cleaning (Python / Pandas)

- Loaded the raw Excel dataset with pandas.
- Checked and removed null values (df.dropna()).
- Removed duplicate records (df.drop_duplicates()).

4.2 Feature Engineering

- Stock_Difference = Optimal_Stock_Level – Demand_Forecast
- Stock_Status = ‘Understock’ if Stock_Difference < 0, else ‘Overstock’
- Priority = ‘High’ for understocked records, ‘Low’ otherwise

4.3 Exploratory Data Analysis (Python / Seaborn / Matplotlib)

- Top 10 drugs by demand — bar chart

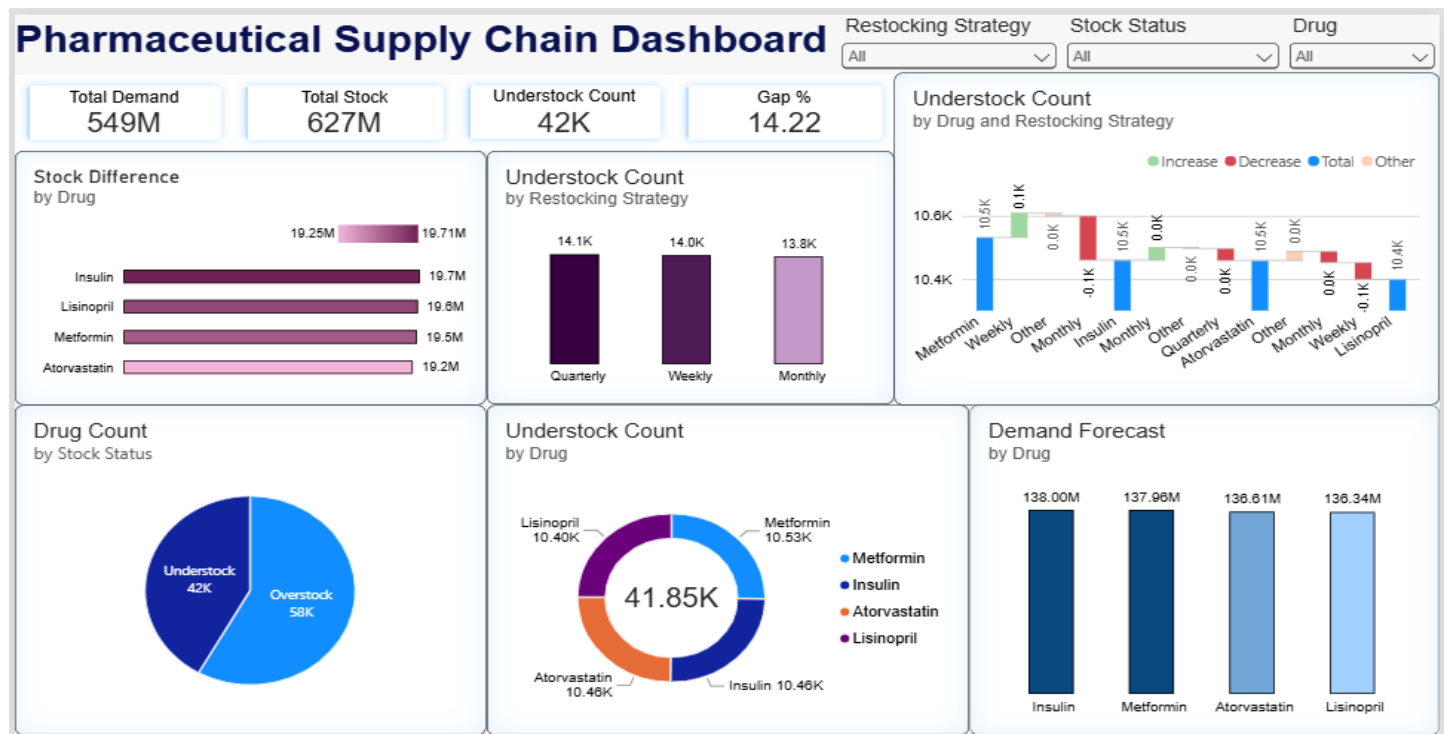
- Stock difference by drug — bar chart
- Understock vs. overstock split — pie chart
- Restocking strategy distribution — count plot
- Demand vs. stock level, colored by stock status — scatter plot
- Demand distribution — histogram with KDE
- Restocking strategy vs. stock status — stacked bar chart

4.4 SQL Analysis (MySQL)

The cleaned dataset was loaded into a MySQL table (supply_data) via SQLAlchemy, and 20 business questions were answered directly in SQL — covering aggregate demand/stock totals, understock/overstock counts and percentages, top critical and top overstocked drugs, strategy-level risk comparison, demand and stock-difference bucketing, and window-function rankings (DENSE_RANK) of drugs by demand and by shortage severity.

4.5 Dashboarding (Power BI)

Findings were consolidated into an interactive Power BI report (Pharmaceutical_Supply_Chain.pbix) for stakeholders to filter by drug and restocking strategy and monitor stock health on an ongoing basis.



5. Key Insights

5.1 Portfolio-level balance

Metric	Value
Total demand forecast	548,911,959 units
Total optimal stock	626,981,930 units
Net position	Overstocked by ~78.1M units in aggregate
Average demand per record	5,489 units
Demand range	1,000 – 9,999 units

5.2 Understock vs. Overstock

- 58.15% of Stock Keeping Unit records (58,145) are Overstocked; 41.85% (41,850) are Understocked.
- Total overstock volume is 196.8M units vs. total understock (shortage) volume of 118.7M units — overstocking is the larger volume problem, but understocking touches a large share of individual Stock Keeping Units and carries higher patient-safety risk.
- Only 173 records (0.17%) land within ± 10 units of optimal — precision stock-setting is rare; most Stock Keeping Units are meaningfully over- or under-planned.

5.3 Restocking strategy has little effect on shortage risk

Strategy	Total Items	Understock %	Avg Stock Difference
Monthly	33,037	41.86%	+787.4
Quarterly	33,522	41.96%	+776.4
Weekly	33,436	41.74%	+778.6

The understock rate is within ~ 0.2 percentage points across all three cadences. This indicates the restocking frequency itself is not the lever driving shortages — the root cause sits upstream, in how Optimal_Stock_Level is forecasted relative to demand, regardless of how often stock is replenished.

5.4 Drug-level exposure is fairly even

Total demand is closely balanced across the four drugs (Insulin 138.0M, Metformin 138.0M, Atorvastatin 136.6M, Lisinopril 136.3M), and average stock difference per drug ranges narrowly from +772 to +788 units. No single drug is disproportionately mismanaged; shortage risk is spread evenly across the portfolio rather than concentrated in one product.

5.5 Critical shortage volume

- 41,850 records show demand exceeding planned stock — these are the records requiring immediate reorder attention.
- Ranking by Stock_Difference (ascending) surfaces the specific highest-shortage Stock Keeping Units for prioritized action.

6. Recommendations

- Prioritize forecasting accuracy over restocking frequency — since cadence does not move the understock rate, effort is better spent improving how Demand_Forecast and Optimal_Stock_Level are calculated.
- Set a tiered safety-stock buffer for the $\sim 41.85\%$ of Stock Keeping Units that run understocked, especially for Insulin given its clinical criticality.
- Investigate the overstock side (58.15% of records, ~ 196.8 M excess units) for working-capital and expiry-risk reduction opportunities.
- Operationalize the SQL queries and Power BI dashboard as a recurring (e.g. weekly) monitoring report rather than a one-off analysis.
- Extend the dataset with unit cost and shelf-life fields in future iterations to quantify the rupee/dollar impact of over- and under-stocking, not just unit counts.

7. Tech Stack

Layer	Tools
Data storage / source	Excel (.xlsx)
Data cleaning & EDA	Python — Pandas, Matplotlib, Seaborn (Jupyter Notebook)
Database & querying	MySQL, SQLAlchemy
Dashboarding	Power BI (.pbix)

8. Conclusion

The pharmaceutical supply chain dataset reveals a portfolio that is net overstocked, yet still leaves over 41% of Stock Keeping Units exposed to shortage — a mismatch driven by forecasting rather than restocking cadence. The SQL, Python, and Power BI deliverables built in this project give stakeholders a repeatable way to monitor this balance and act on the highest-risk drugs and Stock Keeping Units going forward.